AI & SOCIETY COURSE

Prof. Mor Naaman

Part 4

CHALLENGES FOR AI IMPLEMENTATION

Unit 4: The economics of AI

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The move: public data becomes a private good

What went in

Our posts, photos, books, articles, code, and conversations.

Collected at almost no cost — and, as we saw, mostly without consent.

What comes back

The same knowledge, repackaged as a subscription, an API call, or an ad slot.

A shared commons became a private product that we need to pay for (in most cases)

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The result: the most valuable companies on earth

Nvidia	Valuation: ~$4.8 trillion. Sells the chips (almost) every model runs on.
Alphabet/Google, Microsoft	~$3.8T and ~$3.0T. Sell the compute the models run in, and offer their own AI implementations
Amazon, Meta, Broadcom	$1.5–2.3 trillion each. Infrastructure, custom chips, and their own models.
Anthropic, OpenAI	Valuation of trillions, and still private companies (not traded in the market)
Domination:	Six of the ten most valuable public companies in the world are AI companies.

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This unit: goal

- Understand how the AI companies make money
- Understand what are the AI companies' costs

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How do they actually make money?

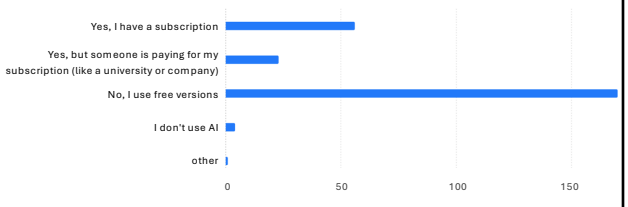
this is where most of the income is coming from

Subscriptions	~900M weekly ChatGPT users, but only a low %% pay; many use the free tier
Enterprise users	Company gives chat access to all employees; data kept confidential
API/Tokens	Use of AI via API calls is counted using "tokens".
Ads (coming next!)	The free majority is increasingly being monetized using ads shown to users.

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Your data (254 responses)



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techradar.

OpenAI says it's making $2 billion a month - mostly from enterprise users

Craig Hale
April 1, 2026 • 2 min read

When yo
syndicat

Anthropic makes money by selling Claude by the token — about 80% of its $65B annualized run-rate revenue comes from the API and enterprise contracts, not consumer subscriptions.

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How do tokens work?

API: a connection that lets your application send a request to a computer and get a response, all done via computation.

Server: a computer on the Internet hosting a service, for example an AI model, responding to API requests.

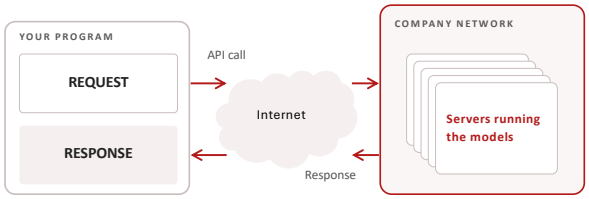
Token: a chunk of text, about 4 characters or ¾ of a word. Models read and write in tokens, and you are billed per token.

Input tokens: the prompt you send to the server's API; **Output tokens:** the text the model writes back.

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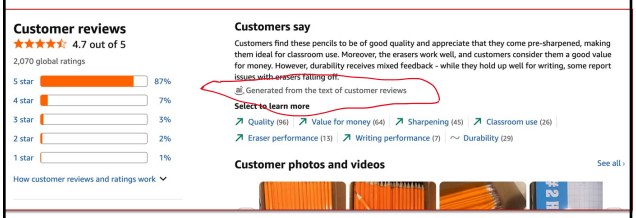
AI model calls via an API



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Tokens: Example Task



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Tokens Example (Sep 2026)

Task: a script summarizes 2,000 customer reviews into one paragraph.
250 input tokens each → 500,000 input; 200 output.

Model tier	Input $/1M	Output $/1M	Total cost
Cheap small model (InclusionAI Ling-3.0-flash)	$0.021	$0.063	**$0.011**
Cheaper frontier (deepseek-v4-pro)	$1.32	$3.96	**$0.66**
Frontier (Claude Fable 5.1)	$15.00	$50.00	**$7.51**

Math (deepseek-v4-pro): 0.5M × $1.32 + 0.0002M × $3.96 = $0.66 + $0.001 ≈ $0.66

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Tokens economics: gets complicated

- Cache hit/miss, off-peak hours discount, batch discounts, …
- **More significantly:**
  - Is the model capable of doing the task well? Need more input/output, or test with smaller models?
  - Reasoning/thinking tokens are billed as output; could be 5–20 times than the final answer!

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Tokens economics: remember the pelican?

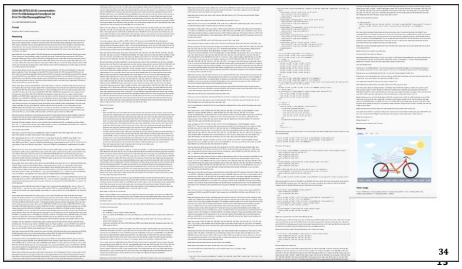
Input:

31 tokens

Output:

13,041 reasoning
2,552 response

15,593 tokens



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Example from my work: API use

What counts as "suspicious shift in profile description" on social media?

Identifying accounts takeover that may suggest scammers, spammers, or other bad players.

Problem: 100,000s profile changes a month!

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Example from my work: API use

What counts as "suspicious shift to profile description" on social media?

MO **Maya Ortega**
@mayaortega_

PhD student in marine biology. Tide pools, bad coffee, worse puns. Boston.

Profile: March 2026

MO **Maya Ortega**
@mayaortega_

Crypto & forex mentor. Financial freedom in 90 days. DM for signals. Not investment advice.

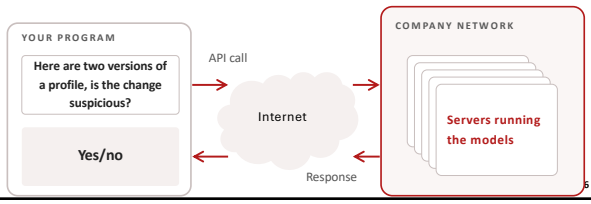
Same account: September 2026

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Example in my work: API use

What counts as "suspicious shift to profile description" on social media?
Wrote a program that sends two versions (before, after) of each profile to a model using an API call.



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Example from my work: API use

Some edits are not a problem – no need to use a very capable model

MO **Maya Ortega**
@mayaortega_

PhD student in marine biology. Tide pools, bad coffee, worse puns. Boston.

Profile: March 2026

MO **Maya Ortega**
@mayaortega_

PhD student in marine biology, writing my dissertation on white sharks.

Same account: September 2026

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Example from my work: API use

Some edits are not a problem – no need to use a very capable model

First pass cheaper model: is the difference between the two profile "small change" "complicated change"? (40,000 requests a day; ~$1 with cheap model)

Second pass frontier model: all the "complicated changes" (100 a day; ~$0.05 with expensive model)

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Summary so far

Models are priced by monthly fee or API/token access

Tokens: huge variability in cost, based on what model you want to use

Most of the AI companies' income is enterprise and token use!

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Expenses: costs for the AI companies?

Training: one frontier model is a huge project with tens of thousands of chips running for months, human annotation, etc.

Inference: every single answer costs money for the AI company to produce (compute costs). When there are trillions of queries, the costs add up.

Infrastructure: building data centers and other projects to be able to support training + inference. Over $700 billion in 2026.

Data agreements: paying publishers and platforms for training data (a few billion dollars industry-wide; a small amount compared to the rest).

Source: CNBC, February 2026 (Alphabet, Amazon, Meta, Microsoft 2026 capex guidance); later estimates run $725–760bn.

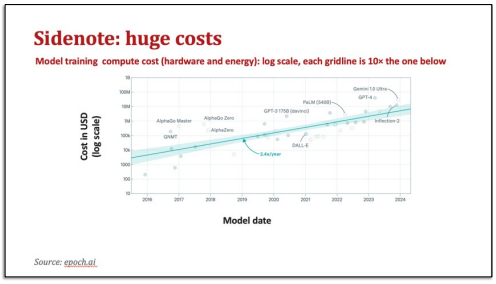
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Remember this?

Sidenote: huge costs

Model training compute cost (hardware and energy): log scale, each gridline is 10× the one below



Source: epoch.ai

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Summary: costs

- Hundreds of millions for "frontier" model training…
  - But can be down to several millions with new techniques for smaller but strong models
- Inference costs is variable with use
- But infrastructure needs are huge: need enough compute ready to to support both model training and inference
- And don't forget data acquisition

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Cutting costs: two approaches

Better engineering

Model training: better algorithms, shortcuts e.g. distillation, ..

Inference: caching, batching, routing, other optimizations

Infrastructure: cheaper/faster chips, optimized server farms, …

External factors

Government subsidies, tax breaks

Negotiate cheaper rates for electricity, water, land

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So what is all of this actually for?

- What do AI companies sell?
  - Not selling better chatbots!
  - Labor replacement: work now done by people, done instead by capital.
- Classic industrial capitalism: technology replacing human wages. But this time: knowledge work.
- Money flows to whoever owns the model and the data center (in the U.S., India, or China?). Not workers, not local companies.
  - More on that later!

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Are these companies sustainable?

- The costs for the companies are steep, and competition is growing
- Multiple AI companies are valued with over 1 Trillion dollars
- Can they justify the valuable? Can they make billions more than they spend?

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Which vision do the 2026 numbers support?

- Anthropic: $65B a year in sales, valued near $965B, hoping for a $2T valuation for the IPO in 2026.
- OpenAI: $40B a year in sales, valued at $852B, still losing money.
- Both are priced at about 15 to 21 times what they earn today.
- Do these valuations make sense – can they keep their sales/revenue growing that fast?

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V1: To stay as profitable as they promise, AI companies need to replace many human jobs

- Building and running the data centers costs hundreds of billions of dollars.
- If AI is replacement for current software: all the world's software spending is too small to pay that back.
- A $20 subscription is too little to pay for all these costs, and token prices are falling because of competition.
- So what the AI companies really have to do to justify their valuation is to replace a lot of human jobs.

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V2: Maybe not! Maybe the AI companies can do well and no jobs need to disappear

- Making every worker slightly faster is already worth trillions of dollars to companies.
- Companies spend more on software with each new technology, not less.
- Cheap AI creates useful work that nobody could pay for before.
- Every new generation of chips makes serving customers cheaper, so it is ok if competition drives price down.

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Summary: AI economics

Public in, private out: Data collected freely from all of us is sold back to us as a service by companies that collected huge evaluations.

Three revenue streams: Subscriptions (private and enterprise), tokens/API use, and (increasingly) advertising.

Token/API economics: Most access to models done via API, and counted in tokens. Huge variability in costs between models.

Costs are huge, but shifting: Compute costs for model training, inference, infrastructure. There is a fight to reduce them, but who is paying?

The goal is labor: Whoever owns the model and the data center captures the value of the work.

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